A Lightweight Multistream Framework for Salient Object Detection in Optical Remote Sensing

Zhenxin Ai, Huilan Luo, and Jianqin Wang

***Abstract*— Salient object detection (SOD) in optical remote sensing images (ORSIs) is challenging due to small object sizes, low contrast, and complex backgrounds. Existing methods often rely on computationally intensive architectures, limiting their efficiency in real-world applications. To address this, we propose LiteSalNet, a lightweight deep learning framework for ORSI SOD. LiteSalNet employs MobileNetV2 as a compact encoder and enhances multiscale feature representation through three modules: the adaptive spatial attention module (ASAM) for spatial attention (SA) refinement, the dual-scale feature enhancement module (DSFEM) for local–global feature integration, and the semantic context enhancement module (SCEM) for high-level semantic refinement. Additionally, a multistream progressively decoding framework (MSPDF) is introduced to decode saliency, edge, and skeleton maps in a supervised manner, improving boundary precision, suppressing background noise, and enhancing internal object consistency. Extensive experiments on two benchmark ORSI datasets demonstrate that LiteSalNet outperforms 19 state-of-the-art (SOTA) models across multiple evaluation metrics, including F-measure (F-m), S-measure, E-measure, and mean absolute error (MAE). Notably, LiteSalNet achieves these results with only 3.90 M parameters and 7.35 G floating-point operations per second (FLOPs), ensuring high computational efficiency. The code and results are available at https://github.com/ai-kunkun/LiteSalNet.**

***Index Terms*— Edge-skeleton integration, hybrid attention mechanism, lightweight networks, multiscale feature learning, optical remote sensing images (ORSIs), salient object detection (SOD).**

I. Introduction

SALIENT object detection (SOD) in optical remote sensing images (ORSIs) aims to identify visually significant objects or regions in complex aerial or satellite imagery [1], [2], [3], [4], [5], [6]. This task is essential for various remote sensing applications, including land-cover classification, urban planning, and disaster monitoring. While SOD has been extensively studied in natural scene images (NSIs), SOD in ORSIs remains challenging due to the unique characteristics of remote sensing data [7], [8], [9], [10], [11], [12].

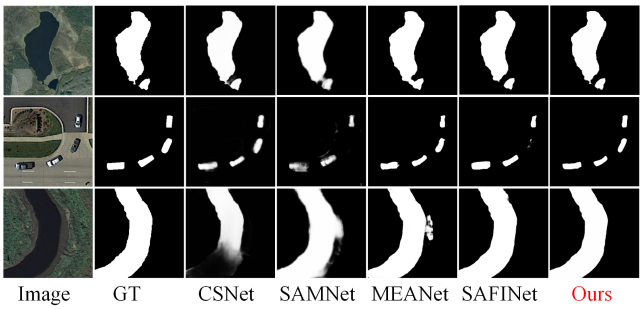

Fig. 1. Challenges in ORSI-SOD: Comparisons between NSI-SOD methods (CSNet [2] and SAMNet [4]) and ORSI-SOD models (MEANet [3] and SAFINet [5]).

ORSIs, captured from high altitudes by satellites or drones, cover large areas with diverse objects, such as buildings, roads, and farmlands. These objects often appear small, have low contrast, and blend into complex backgrounds [13], [14], [15], [16], complicating detection. Additionally, variations in imaging resolutions, geographic distributions, and environmental factors, such as illumination changes and atmospheric interference further challenge reliable SOD, necessitating robust detection models.

Fig. 1 highlights several challenging ORSI-SOD scenarios by comparing NSI-SOD methods (e.g., CSNet [2] and SAMNet [4]) with specialized ORSI-SOD models, such as multiscale edge-embedded attention network (MEANet) [3] and spatial attention feedback iteration network (SAFINet) [5]. These comparisons reveal the limitations of directly applying NSI-SOD models to ORSIs, emphasizing the need for tailored architectures designed explicitly for ORSI-SOD tasks.

Early ORSI-SOD approaches relied on traditional computer vision techniques, such as threshold-based segmentation, region-growing algorithms, and handcrafted features based on contrast, texture, and edges. However, these methods lacked generalization capability and struggled with complex scenes involving varying object appearances, background clutter, and illumination conditions.

With the rise of deep learning, significant progress has been made through automatic feature extraction and end-to-end learning [17], [18]. Convolutional neural networks (CNNs), particularly encoder–decoder architectures, such as U-Net and fully convolutional networks (FCNs), have become popular due to their ability to extract multilevel contextual and spatial

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features [19], [20]. Recent models have further incorporated attention mechanisms, multiscale feature aggregation, and edge-enhanced learning strategies, achieving state-of-the-art (SOTA) performance [3], [5], [21], [22].

Despite these advances, three critical challenges persist in ORSI-SOD. The first challenge is multiscale and contextual feature extraction. Objects in ORSIs appear at various scales due to different altitudes and imaging resolutions. Small objects, such as vehicles or buildings are easily missed, while large-scale objects, such as farmlands require global contextual understanding. Many existing models struggle to combine local and global features effectively, limiting their ability to detect small and large objects simultaneously [23].

The second challenge is incomplete object representation due to single-stream decoding. Many existing models rely on single-stream decoding, focusing only on saliency maps while ignoring other critical geometric features, such as object edges and skeletons. This results in inaccurate boundary localization, internal feature loss, and higher background noise sensitivity. Incorporating edge and skeleton maps can provide complementary structural information, thereby improving saliency prediction [3], [24].

The third challenge is the tradeoff between computational complexity and detection accuracy. Advanced ORSI-SOD models achieve high detection accuracy but often require millions of parameters and extensive floating-point operations per second (FLOPs), making them impractical for real-time applications, such as UAV-based monitoring or onboard satellite processing. Conversely, lightweight models reduce computational complexity but at the cost of lower detection accuracy [4], [6], [25], [26].

Fig. 2 illustrates this tradeoff by comparing models, such as EMFINet [23] and RRNet [27], which achieve high detection accuracy but have high computational demands, against lightweight models, such as SAMNet [28] and CSNet [2], which reduce computational complexity but compromise detection accuracy.

To address these limitations, we propose LiteSalNet, a lightweight and efficient network specifically designed for ORSI-SOD. LiteSalNet is built on the MobileNetV2 backbone [29], enabling multiscale feature extraction with low computational overhead. These features are further refined using three novel modules: the adaptive spatial attention module (ASAM), the dual-scale feature enhancement module (DSFEM), and the semantic context enhancement module (SCEM). Each module serves a distinct role in enhancing feature extraction, multiscale representation, and boundary refinement.

The ASAM is applied to low-level feature maps to selectively highlight spatially informative regions, improving both contextual and local spatial details. The DSFEM processes middle-level feature maps by combining local and global contextual features, thereby improving boundary localization and reducing false detections. The SCEM refines high-level semantic features by strengthening deeper contextual representations, enabling more accurate saliency predictions and reducing background noise.

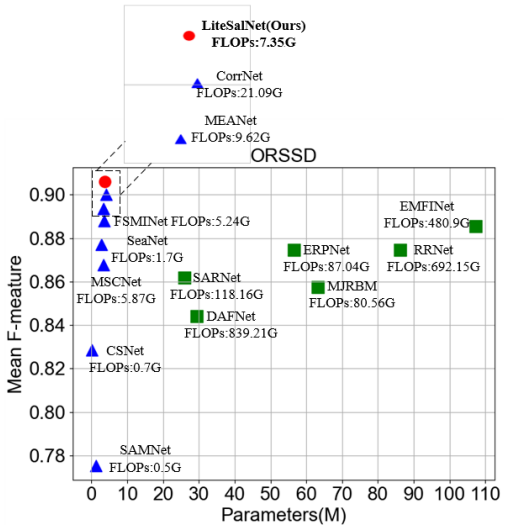

Fig. 2. Comparison of average F-m, parameters, and FLOPs for LiteSalNet versus SOTA methods on the ORSSD dataset. Blue triangles represent lightweight methods, green squares indicate nonlightweight methods, and the red circle denotes LiteSalNet.

Additionally, we introduce the multistream progressively decoding framework (MSPDF), which leverages cross-task and cross-level feature integration to refine saliency predictions. By progressively decoding saliency, edge, and skeleton maps in a supervised manner, MSPDF enhances both boundary precision and internal consistency, enabling robust SOD in complex ORSI environments.

The main contributions of this work are summarized as follows.

1) A lightweight and efficient network, LiteSalNet, is proposed for SOD in optical remote sensing images. Three novel modules—the ASAM, the DSFEM, and the SCEM—are introduced to refine multiscale features while maintaining low computational overhead. These modules collectively enhance feature extraction, multiscale representation, and boundary refinement.
2) A novel MSPDF is developed to simultaneously decode saliency, edge, and skeleton maps using supervised learning. This framework improves detection accuracy by enhancing boundary precision, suppressing background noise, and ensuring structural consistency through multitask learning. Its streamlined architecture effectively balances accuracy and efficiency.
3) Extensive experiments are conducted on two benchmark ORSI datasets, ORSSD and EORSSD, demonstrating that LiteSalNet outperforms 16 SOTA models across multiple evaluation metrics, including F-measure (F-m), S-measure, E-measure, and mean absolute error (MAE). LiteSalNet achieves these results with only 3.90 M parameters and 7.35 G FLOPs, highlighting its superior computational efficiency.

II. Related Works

A. Salient Object Detection

SOD has evolved from early models relying on low-level features, such as color, brightness, texture, and boundary cues.

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Traditional methods like the spectral residual (SR) model [30] used image frequency analysis for saliency detection, while the global contrast (GC) model [31] emphasized feature contrast. A comprehensive review [32] highlighted the limitations of these models in handling complex scenes.

The advent of CNNs transformed SOD by enabling automatic multiscale feature learning. Deeply supervised saliency (DSS) [33] introduced deep supervision to enhance multiscale feature representation. The pyramid spatial attention network (PSANet) [34] used a pyramid spatial attention (SA) mechanism for better context fusion, while the squeeze-and-excitation network (SENet) [35] applied channel attention (CA) for feature enhancement. The convolutional block attention module (CBAM) [36] combined spatial and CA, and nonlocal inception 3-D (NLI3D) [37] extended attention with global context modeling through self-attention mechanisms.

Boundary-aware learning has played a critical role in improving SOD. Edge-aware saliency detection (EASD) [38] introduced an edge-enhanced loss function, while contour-aware saliency detection (CASD) [39] leveraged geometric object boundaries. An aggregated multilevel features network (AMulet) [40] emphasized multilevel feature integration with boundary preservation, and the boundary-aware saliency network (BASNet) [41] applied boundary-aware supervision for precise edge localization.

Recent models have further refined SOD with advanced architectures. The pyramid pooling module network (PoolNet) [42] incorporated edge guidance, while the edge guidance network (EGNet) [43] enabled efficient detection using an interactive two-stream decoder (ITSD). ITSD [44] improved performance through global context-aware progressive aggregation. The global context-aware progressive aggregation network (GCPANet) [45] enhanced results with channel-wise attention in a stacked U-shaped network. The stacked U-shape network with channel-wise attention (SUCA) [46] advanced segmentation using a progressive architecture for global localization, local segmentation, and knowledge review.

B. Optical Remote Sensing Image SOD

Optical remote sensing image SOD (ORSI-SOD) has gained considerable attention due to its applications in geospatial monitoring, land-cover analysis, and disaster management. Early CNN-based methods laid the foundation for this field. LVNet [47] introduced a two-stream pyramid structure with nested V-shaped modules, enabling multiscale object detection by capturing both fine-grained and large-scale features.

Subsequent advancements integrated edge-aware learning for improved boundary precision. A multitask network [48] jointly predicted saliency and edge maps, enhancing object localization. Edge-aware supervision was further extended through multiscale feature extraction across different resolutions, leading to better boundary preservation [49].

The semantic-guided attention refinement network (SARNet) [50] improved saliency detection by leveraging semantic-guided attention refinement. The dense attention fluid network (DAFNet) [51] used dense attention mechanisms for more robust feature learning. Multiscale joint region and boundary model (MJRBM) [52] combined region- and boundary-based models for multiscale feature integration, while edge-guided recurrent positioning network (ERPNet) [53] employed edge-guided recurrent positioning for accurate object detection.

These advancements demonstrate a shift from traditional handcrafted features to deep learning-based models integrating attention mechanisms, multiscale feature aggregation, and edge-aware strategies. However, balancing detection accuracy and computational efficiency remains challenging, particularly for real-time applications, such as UAV-based monitoring and onboard satellite processing.

C. Lightweight SOD

To meet the demands of real-time applications and embedded platforms, lightweight SOD models prioritize reduced parameter counts and computational costs through efficient network designs. Core techniques include depthwise separable convolutions, model pruning, and knowledge distillation. MobileNet [54] and MobileNetV2 [29] pioneered depthwise separable convolutions, significantly reducing model complexity while maintaining performance. Kernel pruning methods [55] removed redundant filters to improve efficiency, while knowledge distillation [56] allowed compact student models to learn from larger teacher models, enabling multiscale feature fusion and edge supervision [57].

In ORSI-SOD, lightweight models have followed similar trends. CorrNet [58], the first lightweight CNN-based ORSI-SOD model, simplified VGG-16 [59] using a coarse-to-fine feature extraction strategy, reducing parameters to 4.09 M but retaining a high computational cost of 21.1 G FLOPs. Subsequent models adopted MobileNetV2 as a lightweight backbone. The multiscale context network (MSCNet) [4] integrated multiscale context information within a lightweight framework. SAFINet [5] improved efficiency using SA feedback, while MEANet [3] introduced multiscale edge embedding, achieving strong performance with only 3.27 M parameters and 9.62 G FLOPs.

D. Multistream Framework

To better exploit edge and boundary information, researchers have developed various multistream frameworks for SOD. ADSTNet [60] introduced a dual-stream network that combines multiscale features for efficient saliency detection in remote sensing images. The pyramid attention edge network (PAENet) [61] employed a pyramid attention structure integrated with a salient edge detection module to enhance object boundary localization. Edge-enhanced dual-stream network (EEDSN) [62] processed edge features in one stream while capturing regional features in the other, improving saliency prediction through complementary feature learning.

Further advancements integrated global and local information through multilevel feature fusion. The edge-aware global fusion network (EAGFN) [63] extracted edge features and fused them with multilevel saliency features to improve detection accuracy. The multistage dual-path network (MSDPN)

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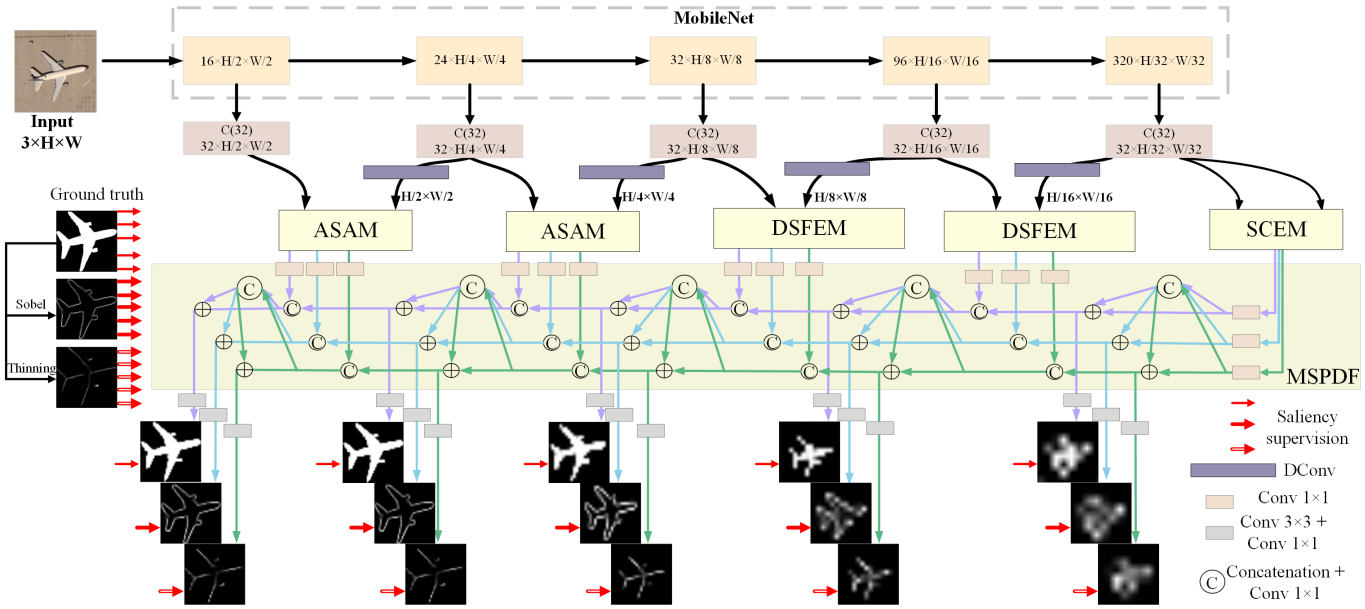

Fig. 3. Overall architecture of LiteSalNet. The network follows an encoder–decoder structure based on MobileNetV2, featuring three specialized modules: ASAM, DSFEM, and SCEM. These modules refine multiscale features, which are further processed through the MSPDF. The MSPDF employs three parallel decoding streams for saliency prediction, edge detection, and skeleton estimation, enabling accurate, boundary-aware, and structure-consistent saliency detection.

[64] separately estimated salient edges and regions, merging fine-grained edge details with global contextual information. The edge-guided three-stream network (EGTSNet) [65] integrated multiscale features through three parallel streams to enhance saliency detection in remote sensing images.

Unlike the aforementioned methods that primarily focus on feature enhancement streams, we propose the MSPDF, which simultaneously decodes saliency, edge, and skeleton maps progressively. This design facilitates lightweight multistream processing, enables deeper feature fusion, improves boundary precision, and ensures structural consistency through joint multitask learning.

III. Methodology

This section presents the proposed LiteSalNet architecture. We begin with an overview of the network in Section III-A, followed by detailed descriptions of its key components: the ASAM in Section III-B, the DSFEM in Section III-C, the SCEM in Section III-D, and the MSPDF in Section III-E. The loss function composition is discussed in Section III-F.

A. Network Overview

As illustrated in Fig. 3, LiteSalNet adopts an encoder–decoder structure with MobileNetV2 serving as a lightweight encoder due to its efficient feature extraction capabilities. The encoder comprises five convolutional blocks that progressively downsample the input image $I \in \mathbb{R}^{3\times H\times W}$ using a stride of 2, producing multiscale feature maps denoted as $f_i \in \mathbb{R}^{c_i\times h_i\times w_i}$, where $h_i = H/2^i$, $w_i = W/2^i$, and channel sizes $c_i = \{16, 24, 32, 96, 320\}$ for $i = 1, \ldots, 5$.

To reduce computational costs, each encoder feature map is passed through a 1×1 convolutional layer to reduce the channel size to 32, followed by two 3×3 convolutional layers. This results in a refined set of feature maps, denoted as $\{s_i \mid i = 1, 2, 3, 4, 5\}$, which are then passed to the decoder for multiscale feature integration and enhancement.

The decoder follows a multibranch structure with five lateral branches that extract and refine spatial, contextual, and semantic information through three specialized modules: the ASAM, the DSFEM, and the SCEM.

The first two branches utilize ASAM to fuse adjacent large-scale feature maps, emphasizing spatial details and preserving high-resolution boundary information. This ensures that critical edge details from lower-level features are retained during decoding.

The next two branches apply DSFEM to integrate middle-level features by combining local and global contextual information. This integration supports precise boundary refinement, accurate object localization, and improved contextual understanding through multiscale feature processing.

The final branch processes the smallest-scale, high-level feature map using SCEM. SCEM refines deep semantic features by suppressing background noise, enhancing contextual representation, and highlighting salient object regions, ensuring accurate global feature refinement for the final saliency prediction.

The highest-level feature map refined by SCEM is fed into the MSPDF. In MSPDF, this feature map is concatenated with features refined by ASAM and DSFEM. The MSPDF processes these combined features through three parallel decoding streams specializing in saliency prediction, edge detection, and skeleton estimation. This multistream decoding approach ensures boundary-aware and structure-aware saliency detection, enhancing overall detection performance through progressive refinement.

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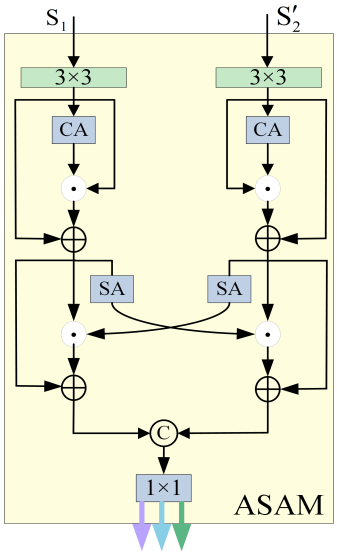

Fig. 4. Detailed structure of the ASAM. The module refines adjacent feature maps through deconvolution and convolution layers, applies CA for feature enhancement, and employs cross-scale SA for context-aware feature refinement.

B. Adaptive SA Module

Balancing spatial precision with semantic richness is crucial for SOD in remote sensing images (RSI-SOD). Shallow, high-resolution features provide detailed spatial information but lack semantic context, while deeper features capture semantic meaning but suffer from reduced spatial resolution. Directly fusing these features can cause spatial blurring and semantic dilution due to resolution mismatch and limited contextual adaptation.

To address this, the ASAM adaptively fuses spatial and semantic features by integrating adjacent feature maps. It selectively transfers contextual information from deeper features to shallower ones while preserving spatial precision, enabling contextually enriched representations for accurate object detection.

As shown in Fig. 3, ASAM is applied twice in a progressive manner. The first instance processes the highest-resolution feature s_1 and its deeper counterpart s_2, while the second refines s_2 with s_3. This hierarchical refinement strategy enhances both shallow and intermediate features for improved saliency prediction.

The detailed ASAM structure is illustrated in Fig. 4. ASAM follows four key steps: feature refinement, CA and residual learning, cross-scale SA, and output fusion, ensuring precise, context-aware feature integration.

1) Feature Refinement: Given adjacent feature maps s_i and s_{i+1}, where $i = 1, 2$, the deeper feature s_{i+1} is upsampled using a deconvolution layer to match the spatial resolution of s_i

$$s'_{i+1} = \mathrm{DConv}_{3\times3}(s_{i+1}). \tag{1}$$

Both features are refined using 3×3 convolutional layers

$$s'_i = \mathrm{Conv}_{3\times3}(s_i), \quad s''_{i+1} = \mathrm{Conv}_{3\times3}\left(s'_{i+1}\right). \tag{2}$$

2) CA and Residual Learning: CA reweights feature channels by emphasizing important features while preserving the original information using residual connections

$$s_i^{\mathrm{ca}} = \mathrm{CA}\left(s'_i\right) \cdot s'_i + s'_i, \quad s_{i+1}^{\mathrm{ca}} = \mathrm{CA}\left(s''_{i+1}\right) \cdot s''_{i+1} + s''_{i+1}. \tag{3}$$

3) Cross-Scale SA: Cross-scale SA allows features from one scale to guide feature refinement at the other scale, enabling mutual feature enhancement

$$\tilde{s}_i = \mathrm{SA}\left(s_{i+1}^{\mathrm{ca}}\right) \cdot s_i^{\mathrm{ca}} + s_i^{\mathrm{ca}}, \quad \tilde{s}_{i+1} = \mathrm{SA}\left(s_i^{\mathrm{ca}}\right) \cdot s_{i+1}^{\mathrm{ca}} + s_{i+1}^{\mathrm{ca}}. \tag{4}$$

4) Output Fusion: Finally, the refined features are concatenated and fused using a 1×1 convolutional layer

$$s_{\mathrm{ASAM}} = \mathrm{Conv}_{1\times1}\left(\mathrm{Concat}\left(\tilde{s}_i, \tilde{s}_{i+1}\right)\right). \tag{5}$$

By processing feature pairs (s_1, s_2) and (s_2, s_3), ASAM enhances spatial precision and semantic richness, enabling robust detection of complex and multiscale salient objects.

C. Dual-Scale Feature Enhancement Module

To aggregate salient features and capture complementary information from adjacent middle-level layers, we propose the DSFEM, as illustrated in Fig. 5. DSFEM integrates features from adjacent deeper layers using two strategies: multiscale feature fusion for local context modeling and cross-attention (CSA) for global context understanding. This dual-mechanism ensures comprehensive feature representation while improving contextual reasoning.

Specifically, features s_i from layers $i \in \{3, 4\}$ are processed by the multireception field attention fusion (Msaf) submodule to enhance semantic information. Simultaneously, the corresponding deeper feature s_{i+1} is upsampled using a deconvolution layer to ensure spatial alignment

$$s'_{i+1} = \mathrm{DConv}_{3\times3}(s_{i+1}). \tag{6}$$

The upsampled feature s'_{i+1} is refined by the dense multiple reception field (Dmrf) submodule. The Msaf output serves as the Query, the Dmrf output as the Key, and the upsampled feature s'_{i+1} as the Value in a CSA mechanism, producing the final enhanced feature representation.

1) Multireception Field Attention Fusion: To accommodate the varying sizes and shapes of salient objects in ORSI, the Msaf module employs multiscale receptive fields through five parallel convolutional branches. This design enables comprehensive feature extraction by capturing contextual information from different spatial extents while maintaining computational efficiency.

The first branch applies a 1×1 convolution for fine-grained local feature extraction, minimizing computational cost while retaining essential details. The remaining four branches utilize a combination of dilated and asymmetric convolutions, expanding the receptive field without significantly increasing the parameter count.

The output of the jth branch is computed as follows:

$$s_{i,j} = \begin{cases} \mathrm{Conv}_{1\times1}(s_i), & j = 1, \\ \mathrm{Conv}_{3\times3,k} \circ \mathrm{Conv}_{k\times1} \circ \mathrm{Conv}_{1\times k}(s_i), & j = 2, 3, 4, 5 \end{cases} \tag{7}$$

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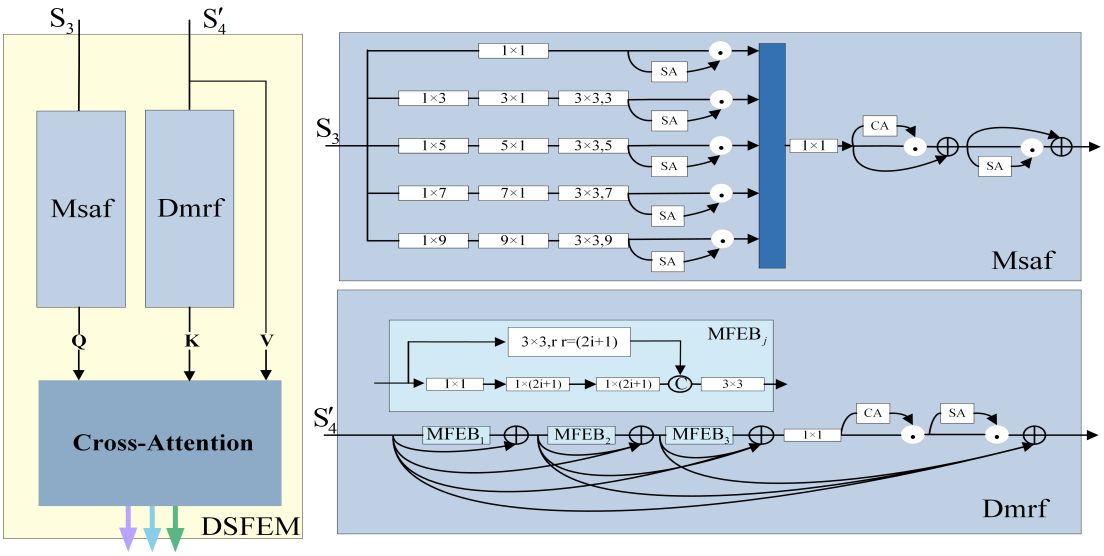

Fig. 5. Structure of the DSFEM module, highlighting its key components: Msaf, Dmrf, and CSA mechanisms. The module leverages multiscale feature extraction, spatial and CA, and CSA refinement for enhanced feature representation.

where $\text{Conv}_{3\times3,k}$ represents a dilated convolution with dilation rate $k = 2j - 1$, while the asymmetric convolutions $\text{Conv}_{k\times1}$ and $\text{Conv}_{1\times k}$ expand the receptive field along horizontal and vertical dimensions. This configuration balances feature diversity and computational efficiency.

To enhance spatial awareness, each branch undergoes SA refinement using the SA operator

$$s_{i,j}^{\text{sa}} = \text{SA}(s_{i,j}) \cdot s_{i,j} + s_{i,j} \tag{8}$$

where $\text{SA}(\cdot)$ computes SA scores to emphasize important spatial features while preserving the original information through residual connections.

The refined features from all branches are concatenated along the channel dimension to form the aggregated feature representation

$$s_{i,\text{concat}} = \text{Concat}\left(s_{i,1}^{\text{sa}}, s_{i,2}^{\text{sa}}, s_{i,3}^{\text{sa}}, s_{i,4}^{\text{sa}}, s_{i,5}^{\text{sa}}\right). \tag{9}$$

The concatenated features are first passed through a 1×1 convolution to reduce the channel dimension, ensuring compact feature representation. The reduced features are further refined using CA as follows:

$$s_{i,\text{ca}} = \text{CA}(s_{i,\text{concat}}) \cdot s_{i,\text{concat}} + s_{i,\text{concat}}. \tag{10}$$

A final SA operation yields the Msaf output

$$f_{i,\text{msaf}} = \text{SA}(s_{i,\text{ca}}) \cdot s_{i,\text{ca}} + s_{i,\text{ca}}. \tag{11}$$

2) Dense Multiple Reception Field Module: As shown in Fig. 5, the Dmrf module extracts multiscale features using three interconnected multiscale feature extraction blocks (MFEB_j), where $j = 1, 2, 3$, following a dense connection strategy. This design promotes feature reuse, reduces redundant computations, and enhances feature representation through cumulative feature aggregation.

Each MFEB employs two parallel branches that balance feature diversity and efficiency. The first branch applies a 3×3 dilated convolution with a dilation rate of $2j + 1$, expanding the receptive field to capture broad contextual information while controlling parameter growth.

The second branch focuses on capturing fine-grained spatial details through a sequence of asymmetric convolutions. This sequence consists of a 1×1 convolution for channel reduction, followed by $1 \times (2j + 1)$ and $(2j + 1) \times 1$ convolutions to expand the receptive field along horizontal and vertical dimensions. This factorized structure increases feature diversity while maintaining computational efficiency compared to a single large convolution.

The outputs of both branches are concatenated along the channel dimension, forming a comprehensive feature representation that combines global and local contexts. To reduce channel dimensionality and avoid feature redundancy, the concatenated features are passed through a 3×3 convolution, compressing the number of channels back to 32.

The final output of the last MFEB_3 undergoes CA and SA to enhance informative features and suppress irrelevant ones. The resulting feature map is then aggregated with intermediate outputs from previous MFEBs through dense connections

$$f_{i+1,\text{dmrf}} = f_{i+1}^{\text{sa}} + \sum_{j=1}^{3} f_{i+1}^{j} + s'_{i+1} \tag{12}$$

where f_{i+1}^{sa} denotes the output of MFEB_3, f_{i+1}^{j} represents the intermediate feature maps from the jth MFEB, and s'_{i+1} is a skip connection from the upsampled feature map. This aggregation reinforces feature propagation, preserves spatial details, and ensures robust multiscale representation.

3) CSA Mechanism: The CSA mechanism establishes long-range contextual relationships between cross-scale feature maps. Conventional convolutions are constrained by fixed receptive fields, limiting their ability to capture global dependencies. In contrast, CSA dynamically computes attention scores between feature representations from different scales, allowing for more comprehensive contextual reasoning.

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The Query, Key, and Value representations are derived from the output features of preceding modules. Specifically, the output of the Msaf module $f_{i,\text{msaf}}$ is used as the Query, the output of the Dmrf module $f_{i+1,\text{dmrf}}$ serves as the Key, and the upsampled feature map s'_{i+1} is treated as the Value

$$Q = W_Q \cdot f_{i,\text{msaf}}, \quad K = W_K \cdot f_{i+1,\text{dmrf}}, \quad V = W_V \cdot s'_{i+1} \tag{13}$$

where W_Q, W_K, and W_V are learnable projection matrices that transform the feature maps into a unified embedding space for attention computation.

The CSA output is computed using scaled dot-product attention, which measures feature similarity across scales

$$f_{\text{DSFEM}} = \text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^{\text{T}}}{\sqrt{d}}\right)V \tag{14}$$

where d denotes the feature dimension, ensuring numerical stability through scaling. The attention scores $\text{Softmax}(QK^{\text{T}}/\sqrt{d})$ indicate the relevance between features from different scales, enabling the selective transfer of contextual information.

By dynamically aggregating multiscale features, the CSA mechanism enhances long-range contextual reasoning, facilitating more accurate and context-aware feature integration.

D. Semantic Context Enhancement Module

In U-Net-based architectures, such as LiteSalNet, high-level semantic features progressively passed down to shallower layers often suffer from semantic dilution, resulting in incomplete target structures or over-predicted foreground regions in the final saliency maps. This issue arises because the high-level semantic features lose their discriminative power during the decoding process, weakening the network's ability to accurately localize salient regions.

To mitigate this limitation, we propose the SCEM, designed to enhance the highest-level semantic feature s_5 by applying attention mechanisms and global pooling operations. The SCEM preserves and amplifies critical high-level semantic information before integration with shallower layers, ensuring more precise saliency prediction. Specifically, the SCEM strengthens feature representations using both channel and SA mechanisms, while a global pooling operation captures broader contextual cues from the highest-level feature map.

As illustrated in Fig. 6, the high-level semantic feature s_5 is processed through global max-pooling (GMP) and a Sigmoid activation to generate a global activation map, emphasizing important regions

$$s_5^g = \text{Sigmoid}(\text{GMP}(s_5)), \quad s'_5 = s_5 \odot s_5^g. \tag{15}$$

The enhanced feature s'_5 undergoes a 1×1 convolution for refinement

$$s''_5 = \text{Conv}_{1\times1}(s'_5). \tag{16}$$

CA and SA are sequentially applied to adaptively re-weight feature channels and enhance spatially important regions

$$s_5^{\text{ca}} = \text{CA}(s''_5) \odot s''_5, \quad s_5^{\text{sa}} = \text{SA}(s_5^{\text{sa}}) \cdot s_5^{\text{ca}}. \tag{17}$$

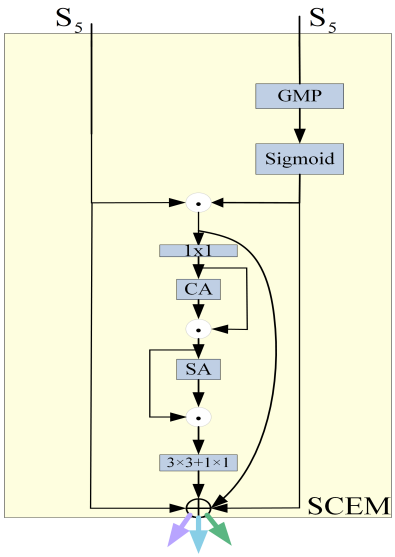

Fig. 6. Illustration of the proposed SCEM architecture, showing the integration of global and local semantic enhancements through attention mechanisms and convolutional layers.

The spatially refined feature s_5^{sa} is processed by a combination of 3×3 and 1×1 convolutions for spatial structure refinement and channel adjustment

$$s_5^{\text{enh}} = \text{Conv}_{1\times1}(\text{Conv}_{3\times3}(s_5^{\text{sa}})). \tag{18}$$

Finally, the original feature, global activation map, and refined features are aggregated via element-wise addition

$$f_{\text{SCEM}} = s_5^{\text{enh}} + s_5 + s_5^g + s'_5. \tag{19}$$

The final output f_{SCEM} combines global context and local feature refinement, enhancing both semantic and spatial representations for accurate saliency detection.

E. Multistream Progressively Decoding Framework

The proposed decoding framework employs a multistream progressive refinement strategy, consisting of three parallel feature refinement streams.

1) *Saliency Stream:* This stream predicts the saliency map by capturing global contextual information while preserving fine details.
2) *Edge Stream:* This stream enhances boundary features, enabling precise object contour detection through explicit edge supervision.
3) *Skeleton Stream:* This stream captures the central structure of salient objects, promoting internal consistency by preserving the overall shape of salient regions.

The proposed framework addresses two key challenges: 1) improving boundary precision through the edge stream and 2) ensuring internal structural consistency through the skeleton stream. Each stream iteratively refines its corresponding feature representation by leveraging low-level spatial features from earlier layers, progressively enhancing decoding outputs.

1) Stream Initialization and Supervision: The output of the SCEM module is processed through three parallel 1×1 convolutional layers, producing three feature maps corresponding

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to the salient object map, salient edge map, and salient skeleton map. Ground-truth supervision is provided by corresponding target maps: the salient edge map is computed using the Sobel operator on the salient object map, while the salient skeleton map is obtained using a thinning operation.

2) Interstream Feature Interaction: To enable effective feature exchange among the streams, the three feature maps are concatenated along the channel dimension and passed through a 1×1 convolutional layer to adjust the channel count back to the original size. The resulting feature map is added element-wise to the individual feature maps from each stream as follows:

$$f_{\text{con}} = \text{Conv}_{1\times1}(\text{Concat}(f_{\text{sal}}, f_{\text{edge}}, f_{\text{skel}})) \tag{20}$$

$$f_{\text{sal}}^{\text{ref}} = f_{\text{sal}} + f_{\text{con}}, \quad f_{\text{edge}}^{\text{ref}} = f_{\text{edge}} + f_{\text{con}}, \quad f_{\text{skel}}^{\text{ref}} = f_{\text{skel}} + f_{\text{con}} \tag{21}$$

where f_{sal}, f_{edge}, and f_{skel} denote the feature maps from the saliency, edge, and skeleton streams, respectively, while f_{inter} represents the interstream refined feature map.

3) Progressive Refinement Process: The interstream refinement process is repeated four additional times after integrating the outputs of the two DSFEM modules and two ASAM modules. This progressive decoding mechanism merges earlier feature representations with new features generated by the DSFEM and ASAM modules, enabling deeper network layers to progressively refine predictions for saliency, edge, and skeleton maps.

At each decoding stage, the refined features from the saliency, edge, and skeleton streams are concatenated with the corresponding outputs of the DSFEM or ASAM modules. This hierarchical feature integration progressively improves prediction accuracy as the decoding process advances.

4) Training and Supervision: During training, the refined feature maps $f_{\text{sal}}^{\text{ref}}$, $f_{\text{edge}}^{\text{ref}}$, and $f_{\text{skel}}^{\text{ref}}$ from all five decoding stages are supervised by their respective ground-truth salient object, edge, and skeleton maps. Specifically, each feature map undergoes a sequence of 3×3 and 1×1 convolutional layers to produce final predictions, which are supervised by the corresponding ground-truth targets.

This multistage deep supervision ensures that each stream specializes in distinct aspects of salient object representation while facilitating complementary learning among streams. By combining multiscale contextual information, precise boundary localization, and consistent internal structure representation, the proposed framework effectively generates accurate and complete saliency maps.

During inference, only the salient object prediction from the final decoding stage is used as the model's output.

F. Loss Function

The proposed model's loss function integrates three complementary components: the pixel-wise binary cross entropy (BCE) loss, the map-level intersection over union (IoU) loss, and the metric-aware F-m loss. These terms balance pixel-level accuracy, region-level consistency, and performance-oriented evaluation metrics.

The BCE loss ensures pixel-level classification

$$L_{\text{bce}} = -\frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \left[G_{ij} \log(p_{ij}) + (1 - G_{ij}) \times \log(1 - p_{ij}) \right]. \tag{22}$$

The IoU loss maximizes overlap between predicted and ground-truth regions

$$L_{\text{iou}} = 1 - \frac{\sum_{i=1}^{H} \sum_{j=1}^{W} p_{ij} G_{ij}}{\sum_{i=1}^{H} \sum_{j=1}^{W} \left(p_{ij} + G_{ij} - p_{ij} G_{ij} \right)}. \tag{23}$$

The F-m loss balances precision and recall with emphasis on precision using a weight factor $\beta^2 = 0.3$

$$L_{\text{fm}} = 1 - \frac{(1 + \beta^2) \cdot \text{precision} \cdot \text{recall}}{\beta^2 \cdot \text{precision} + \text{recall}} \tag{24}$$

where precision and recall are defined as follows:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{25}$$

With

$$\text{TP} = \sum_{i=1}^{H} \sum_{j=1}^{W} p_{ij} G_{ij}$$

$$\text{FP} = \sum_{i=1}^{H} \sum_{j=1}^{W} p_{ij} (1 - G_{ij})$$

$$\text{FN} = \sum_{i=1}^{H} \sum_{j=1}^{W} (1 - p_{ij}) G_{ij}.$$

The model is supervised at five decoding stages for saliency, edge, and skeleton prediction, as indicated by the red arrows in Fig. 3. The total loss function is

$$L_{\text{total}} = \sum_{i=1}^{5} \left[L_{\text{bce}}\left(\text{up}\left(P_s^i\right), G_s\right) + L_{\text{iou}}\left(\text{up}\left(P_s^i\right), G_s\right) + L_{\text{fm}}\left(\text{up}\left(P_s^i\right), G_s\right) + L_{\text{bce}}\left(\text{up}\left(P_e^i\right), G_e\right) + L_{\text{bce}}\left(\text{up}\left(P_k^i\right), G_k\right) \right] \tag{26}$$

where G_s, G_e, and G_k are the ground-truth saliency, edge, and skeleton maps, and P_s^i, P_e^i, and P_k^i are the corresponding predicted maps from the ith decoding stage. The function up$(\cdot)$ denotes upsampling to match ground-truth resolution.

IV. Experiments

A. Experimental Setup

The experiments were conducted on two publicly available ORSI-SOD datasets: ORSSD and EORSSD. Model performance was evaluated using four standard metrics: F-m (F_β^{mean}, $\beta^2 = 0.3$), E-measure (E_ξ^{mean}), S-measure (S_m, $\alpha = 0.5$), and MAE. These metrics measure precision–recall (PR) balance, structural similarity, spatial coherence, and pixel-level accuracy, respectively. The first three metrics are higher better, while MAE is lower better.

The implementation was based on the PyTorch framework and ran on an NVIDIA GeForce RTX 2080Ti GPU. Input images were resized and augmented with multiscale scaling

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TABLE I

QUANTITATIVE COMPARISON OF SOD PERFORMANCE ON TWO BENCHMARK DATASETS. THE TOP THREE SCORES FOR EACH EVALUATION METRIC ARE HIGHLIGHTED IN RED (BEST), BLUE (SECOND-BEST), AND GREEN (THIRD-BEST), RESPECTIVELY

Method	Type	Backbone	Params (M)↓	FLOPs (G)↓	Speed (FPS↑)	EORSSD				ORSSD			
						$F_\beta^{mean}\uparrow$	$E_\xi^{mean}\uparrow$	$S_m\uparrow$	MAE ↓	$F_\beta^{mean}\uparrow$	$E_\xi^{mean}\uparrow$	$S_m\uparrow$	MAE ↓
$ITSD_{20}$ [44]	NSI-SOD	VGG16	17.08	54.5	-	0.8221	0.9407	0.9050	0.0106	0.8502	0.9482	0.9050	0.0165
$GCPANet_{20}$ [45]	NSI-SOD	ResNet50	67.06	54.3	15	0.7905	0.9167	0.8869	0.0102	0.8433	0.9341	0.9026	0.0168
$SUCA_{21}$ [46]	NSI-SOD	ResNet50	117.71	56.4	-	0.7949	0.9277	0.8988	0.0097	0.8237	0.9400	0.8989	0.0145
$PA\text{-}KRN_{21}$ [66]	NSI-SOD	ResNet50	141.06	617.7	12	0.8358	0.9536	0.9192	0.0104	0.8727	0.9620	0.9239	0.0139
$LVNet_{19}$ [47]	ORSI-SOD	-	-	-	-	0.7356	0.8826	0.8644	0.0145	0.7995	0.9259	0.8815	0.0207
$SARNet_{21}$ [50]	ORSI-SOD	VGG16	25.91	118.16	32	0.8541	0.9555	0.9240	0.0099	0.8619	0.9477	0.9134	0.0187
$DAFNet_{21}$ [51]	ORSI-SOD	Res2Net-50	29.35	839.21	18	0.7980	0.9382	0.9184	0.0053	0.8442	0.9537	0.9118	0.0106
$MJRBM_{22}$ [52]	ORSI-SOD	ResNet50	63.28	80.56	19	0.8058	0.9212	0.9091	0.0099	0.8573	0.9394	0.9211	0.0146
$ERPNet_{22}$ [53]	ORSI-SOD	VGG16	56.48	87.04	34	0.8304	0.9401	0.9210	0.0089	0.8745	0.9566	0.9254	0.0135
$RRNet_{22}$ [27]	ORSI-SOD	ResNet-50	86.27	692.15	86	0.8377	0.9449	0.9264	0.0074	0.8747	0.9553	0.9339	0.0112
$EMFINet_{22}$ [21]	ORSI-SOD	VGG16	107.26	480.9	17	0.8486	0.9604	0.9290	0.0084	0.8856	0.9671	0.9366	0.0109
$CSNet_{20}$ [2]	L.N.S	-	0.14	0.7	21	0.7656	0.8929	0.8364	0.0169	0.8285	0.9171	0.8910	0.0186
$SAMNet_{21}$ [28]	L.N.S	-	1.33	0.5	27	0.6879	0.8473	0.8537	0.0151	0.7753	0.8930	0.8835	0.0214
$MSCNet_{22}$ [4]	L.R.S	MobileNet V2	3.26	5.87	58	0.8151	0.9551	0.9071	0.0090	0.8676	0.9653	0.9227	0.0129
$CorrNet_{22}$ [58]	L.R.S	-	4.09	21.1	-	0.8620	0.9646	0.9289	0.0083	0.9002	0.9746	0.9380	0.0098
$FSMINet_{22}$ [26]	L.R.S	-	3.56	5.24	-	0.8436	0.9567	0.9255	0.0079	0.8878	0.9672	0.9361	0.0101
$SeaNet_{23}$ [6]	L.R.S	MobileNet V2	2.76	1.7	72	0.8519	0.9651	0.9208	0.0073	0.8772	0.9722	0.9260	0.0105
$MEANet_{24}$ [3]	L.R.S	MobileNet V2	3.27	9.62	42	0.8678	0.9658	0.9282	0.0070	0.8934	0.9730	0.9340	0.0098
$SoLNet_{24}$ [25]	L.R.S	RepVGG-A0	6.5	8.1	-	0.8513	0.9577	0.9171	0.0078	0.8946	0.9689	0.9284	0.0111
Ours	L.R.S	MobileNet V2	3.9	7.35	35	0.8809	0.9634	0.9273	0.0063	0.9056	0.9700	0.9372	0.0090

NSI-SOD: NSI-SOD method; RSI-SOD: ORSI-SOD method; L.N.S: Lightweight NSI-SOD method; L.R.S: Lightweight ORSI-SOD method

factors [0.75, 1.0, 1.25]. The network was optimized using the Adam optimizer with an initial learning rate of 0.0001, reduced by a factor of 10 after 30 epochs. The batch size was set to 8, with 56 training epochs for the EORSSD dataset and 58 for the ORSSD dataset.

B. Performance Comparison

The proposed model was evaluated against 19 SOTA SOD models, including four NSI-SOD methods: ITSD [44], GCPANet [45], SUCA [46], and PA-KRN [66], as well as seven ORSI-SOD methods: LVNet [47], SARNet [50], DAFNet [51], MJRBM [52], ERPNet [53], RRNet [27], and EMFINet [21].

Additionally, the evaluation included two lightweight NSI-SOD models: CSNet [2] and SAMNet [28], along with six lightweight ORSI-SOD models: MSCNet [4], CorrNet [58], FSMINet [26], SeaNet [6], MEANet [3], and SoLNet [25]. Saliency maps were obtained from publicly available repositories provided by Li et al. [58] or directly from the original authors. Consistent evaluation protocols and standardized assessment codes were applied to ensure a fair and reliable comparison.

1) Quantitative Evaluation: Table I presents a quantitative comparison of the proposed model with 16 SOTA methods on two RSI-SOD datasets using four widely adopted evaluation metrics: F_β^{mean}, E_ξ^{mean}, S_m, and MAE. For each metric, the top three scores are highlighted in red, blue, and green, respectively. The symbols "↑" and "↓" indicate whether higher or lower values are preferable.

The results reveal two key observations. First, retraining NSI-SOD models on RSI-SOD datasets often leads to suboptimal performance compared to models specifically designed for remote sensing. This underscores the importance of architectures that address challenges, such as scale variations, complex backgrounds, and low object contrast.

Second, the proposed LiteSalNet consistently outperforms existing approaches, securing the highest rank in three of the eight evaluation metrics and the second-best performance in two others. This demonstrates its superior overall effectiveness.

On the EORSSD dataset, LiteSalNet achieves the best performance in one metric and ranks second in another. Compared to MEANet, the second-best model, LiteSalNet improves F_β^{mean} by 1.31% and reduces MAE by 0.07%, although its E_ξ^{mean} and S_m scores are slightly lower.

Similarly, on the ORSSD dataset, LiteSalNet outperforms all competing models in F_β^{mean} and MAE while securing the second-best S_m score. Although its E_ξ^{mean} score is marginally lower, LiteSalNet consistently surpasses lightweight models, such as CSNet, SAMNet, and MSCNet across all metrics. These results demonstrate its robustness, accuracy, and computational efficiency in complex remote sensing scenarios.

Additionally, Fig. 7 illustrates the PR and F-m curves of 12 ORSI-SOD models, including LVNet [47], SARNet [50], DAFNet [51], MJRBM [52], ERPNet [53], RRNet [27], EMFINet [21], MSCNet [4], FSMINet [26], CorrNet [58], SeaNet [6], and MEANet [3], along with the proposed LiteSalNet. On the EORSSD dataset, the proposed method (depicted by the red curve) consistently achieves superior performance, maintaining proximity to the top-right corner of both PR and F-m curves. On the ORSSD dataset, its F-m curve remains superior to those of all competing models, while its PR curve is slightly outperformed by CorrNet.

These quantitative results demonstrate that the proposed LiteSalNet model is both robust and highly competitive for remote sensing image SOD tasks.

2) Qualitative Analysis: Fig. 8 presents a visual comparison of the predicted results produced by the proposed model and 12 SOTA ORSI-SOD models listed in Table I. These comparisons cover a wide range of challenging ORSI scenarios, including large objects, small objects, multifocus acquisitions, cluttered backgrounds, elongated objects, and low-contrast targets. The first column in Fig. 8 displays the input images, the second column shows the corresponding

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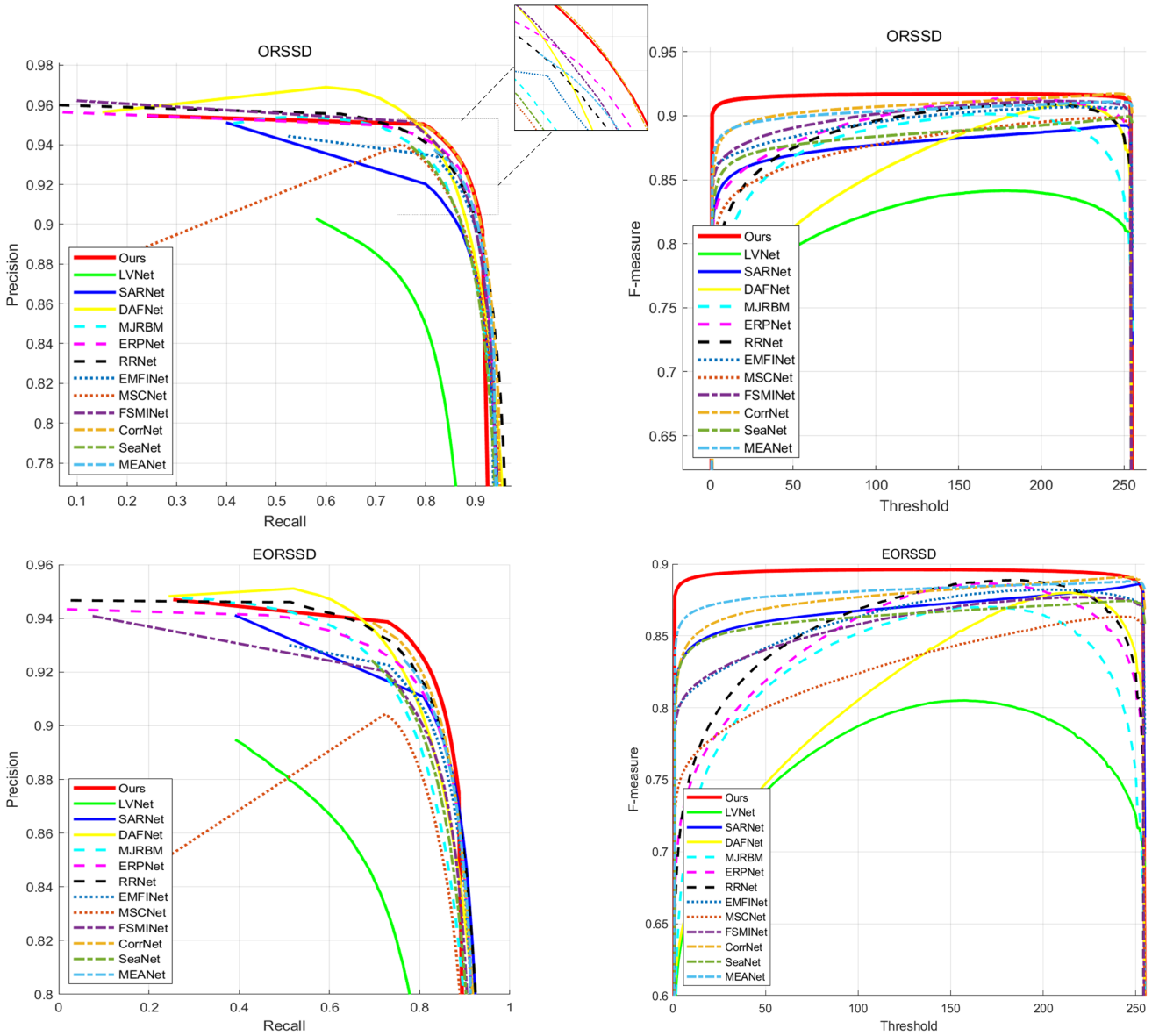

Fig. 7. (Left column) PR curves and (right column) F-m curves on the ORSSD and EORSSD datasets.

ground-truth annotations, while the subsequent 12 columns present the prediction results of competing models, with the final column showing the results produced by the proposed model.

In the first row of Fig. 8, a large elliptical stadium is depicted. Several models, including EMFINet, FSMINet, and CorrNet, incorrectly predicted the salient object as square-shaped due to limited resolution capabilities. In contrast, the proposed model accurately segmented the elliptical stadium, demonstrating its ability to capture fine-grained spatial details.

The third row illustrates a cluttered background scene. Many competing models struggled with adequately segmenting the salient object, often missing critical boundary details. Rows four through six depict scenarios where the salient objects closely resemble the surrounding background due to low contrast. Despite these difficult conditions, the proposed model precisely segmented the complete salient objects, consistently outperforming its counterparts.

Particularly noteworthy is the sixth row, featuring an L-shaped building. Models such as CorrNet, SeaNet, and MSCNet misclassified parts of the building as the background, and some models entirely failed due to the low contrast between the object and its surroundings. In contrast, the proposed model successfully segmented the L-shaped structure with well-defined edges, clearly demonstrating its strong boundary delineation capability.

Overall, the qualitative results in Fig. 8 highlight the robustness and reliability of the proposed model in detecting salient objects across various complex scenes. The proposed method not only ensures accurate edge localization but also effectively preserves the internal structure of salient objects. These results demonstrate the model's ability to propagate spatial information while maintaining critical feature details,

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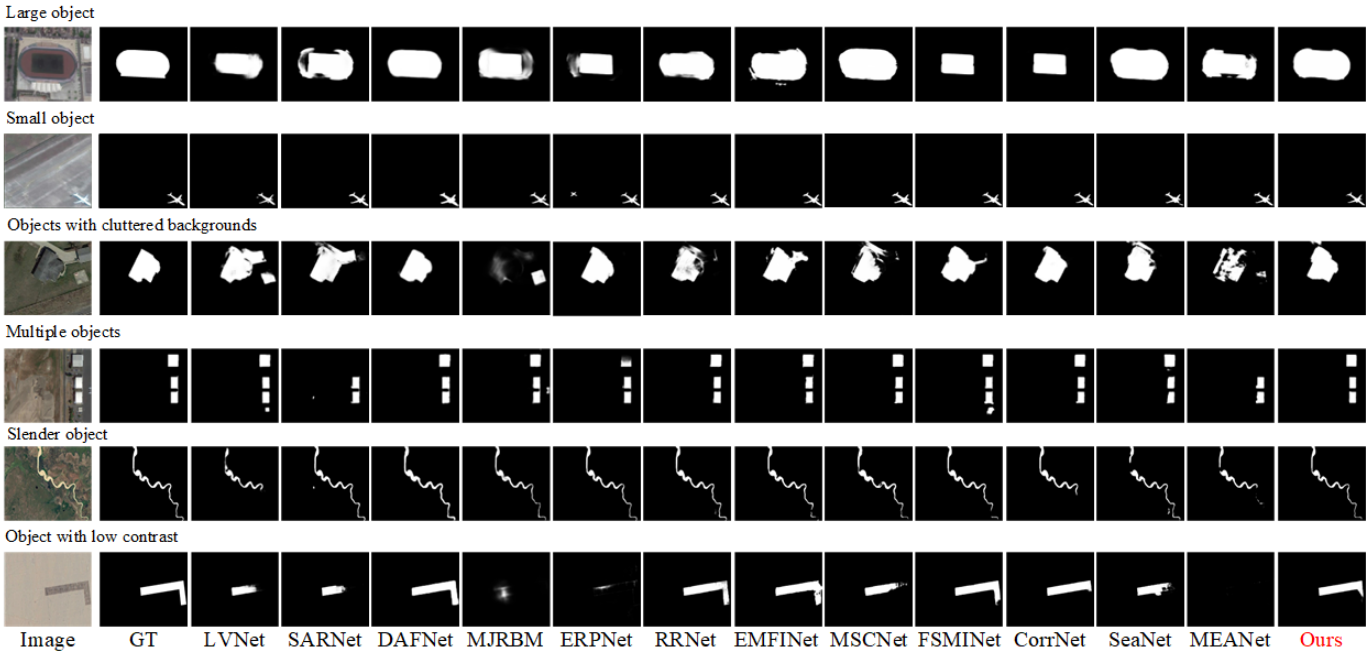

Fig. 8. Visual comparisons with 12 SOTA ORSI-SOD models on various challenging ORSI scenarios. See the zoomed-in view for better visualization.

validating its superior performance in remote sensing image SOD tasks.

3) Comparison of Computational Efficiency: Table I summarizes the computational cost and model complexity of the proposed method relative to SOTA approaches. Notably, the proposed model requires only 3.90 M parameters, which is significantly lower than RRNet (86.27 M) and EMFINet (107.26 M). Additionally, its computational load is only 7.35 G FLOPs, compared to 692.15 G FLOPs for RRNet and 480.9 G FLOPs for EMFINet. These reductions highlight the efficiency of the proposed model without compromising performance.

Compared to lightweight models, such as CSNet, SAM-Net, and MSCNet, the proposed approach incurs a slightly higher parameter count and FLOPs. However, it consistently outperforms these competitors across all eight evaluation metrics on both datasets. This modest increase in computational cost is offset by significant improvements in detection accuracy, establishing a strong balance between efficiency and performance.

To ensure a fair speed comparison in terms of frames per second (FPS), we evaluated the models with publicly available trained weights on an NVIDIA GeForce RTX 2080Ti GPU. The results, presented in Table I, indicate that LiteSalNet achieves an inference speed of 35 frames/s. While this is lower than some other lightweight models, such as MSCNet, SeaNet, and MEANet, it still exceeds the standard video frame rate of 30 frames/s. These findings demonstrate that LiteSalNet not only maintains high detection accuracy but also meets real-time processing requirements, making it well-suited for deployment in resource-constrained environments.

C. Ablation Study

This section presents a series of ablation experiments conducted on the EORSSD dataset to evaluate the contributions of key modules within LiteSalNet. The experiments assess the independent roles of the ASAM, DSFEM, and SCEM modules while analyzing their synergistic effects within the overall architecture.

1) Effectiveness of Key Modules: We conducted ablation experiments on the ASAM, DSFEM, and SCEM modules to evaluate their contributions to the model's performance. As shown in Table II, these experiments validate the unique role of each module while highlighting their complementary effects within the network.

Adding only the ASAM module improved all evaluation metrics, demonstrating that enhancing low-level features strengthens the decoding process by providing detailed spatial information. Similarly, incorporating only the DSFEM module resulted in consistent performance gains, indicating that refining middle-level features improves saliency prediction through enhanced contextual understanding. In contrast, using only the SCEM module led to slight performance drops in three metrics, with only MAE showing a marginal improvement. This suggests that focusing solely on high-level features without refining lower-level features limits spatial precision and boundary clarity.

Combining the SCEM module with either ASAM or DSFEM led to better results compared to using SCEM alone. Integrating all three modules (ASAM, DSFEM, and SCEM) achieved the best overall performance, confirming the importance of enhancing and integrating features across multiple levels within the network.

Furthermore, we analyzed the computational cost of each module by measuring parameter count, FLOPs, and inference speed in FPS, as shown in Table II. The ASAM and SCEM modules introduce minimal computational overhead, adding only 0.03 and 0.02 M parameters, respectively, while DSFEM contributes 0.28 M parameters compared to the baseline. Due to the dense convolution operations applied in DSFEM

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TABLE II

QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS ON THE EORSSD DATASET, EVALUATING THE CONTRIBUTIONS OF KEY COMPONENTS IN LITESALNET. EACH CONFIGURATION REFLECTS THE INCLUSION OR EXCLUSION OF THE ASAM, DSFEM, AND SCEM MODULES, HIGHLIGHTING THEIR INDIVIDUAL AND COMBINED EFFECTS ON MODEL PERFORMANCE

Components				Params (M) ↓	FLOPs (G) ↓	Speed (FPS) ↑	EORSSD			
Baseline	ASAM	DSFEM	SCEM				F_β^{mean} ↑	E_ξ^{mean} ↑	S_m ↑	MAE ↓
✓				3.58	6.53	62	0.8482	0.9452	0.9095	0.0095
✓	✓			3.61	6.92	55	0.8546	0.9589	0.9218	0.0082
✓		✓		3.86	6.96	38	0.8610	0.9493	0.9144	0.0074
✓			✓	3.60	6.53	58	0.8416	0.9355	0.8972	0.0090
✓		✓	✓	3.88	6.96	38	0.8683	0.9533	0.9176	0.0073
✓	✓		✓	3.62	6.93	49	0.8479	0.9558	0.9194	0.0076
✓	✓	✓		3.88	7.35	36	0.8670	0.9535	0.9211	0.0070
✓	✓	✓	✓	3.90	7.35	35	0.8809	0.9634	0.9273	0.0063

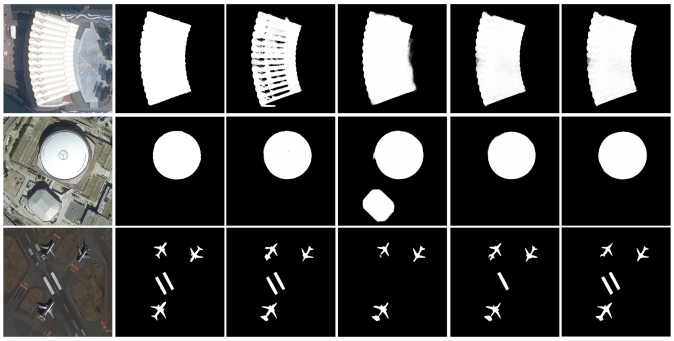

Fig. 9. Visualization of the effectiveness of key modules in LiteSalNet. Each row illustrates different challenging remote sensing scenarios. The first column shows the input images, while the second column displays the corresponding ground-truth annotations. The subsequent columns present the model's prediction results under different ablation settings: without ASAM, without DSFEM, and without SCEM, respectively, followed by the full model.

for multiscale and contextual feature extraction, the FPS drops from 62 (baseline) to 38. These results highlight the tradeoff between accuracy and efficiency when integrating computationally intensive components in ORSI-based saliency detection models.

The visual effectiveness of these modules is illustrated in Fig. 9. Removing ASAM caused a noticeable degradation in boundary clarity and spatial details, especially around the white building in the first row. This underscores ASAM's role in refining low-level features through CA and SA, enabling precise boundary localization and feature integration. Without ASAM, the model struggled with contextual understanding, reducing detection precision and consistency.

Similarly, removing DSFEM led to significant performance drops in middle-level feature processing. As shown in Fig. 9, the edges of the white building in the first row became blurred, detection errors appeared in the second row, and the airplane and boxes in the third row were missed. This highlights DSFEM's role in extracting multiscale contextual features through multireception field attention and dense feature refinement, enabling robust mid-level feature representation.

TABLE III

QUANTITATIVE EVALUATION OF THE IMPACT OF KEY MSPDF COMPONENTS ON MODEL PERFORMANCE ON THE EORSSD DATASET. EACH CONFIGURATION REFLECTS THE INCLUSION OR EXCLUSION OF EDGE STREAM (EDG) AND SKELETON STREAM (SKE) IN THE MSPDF MODULE. PERFORMANCE IS MEASURED USING FOUR WIDELY ADOPTED EVALUATION METRICS. THE BEST SCORES FOR EACH METRIC ARE HIGHLIGHTED IN RED

Components			EORSSD			
Sal	Edg	Ske	F_β^{mean} ↑	E_ξ^{mean} ↑	S_m ↑	MAE ↓
✓			0.8629	0.9596	0.9251	0.0079
✓	✓		0.8709	0.9610	0.9196	0.0070
✓		✓	0.8639	0.9613	0.9262	0.0069
✓	✓	✓	0.8809	0.9634	0.9273	0.0063

The SCEM module adaptively fuses high-level semantic features, enhancing spatial precision and edge clarity in complex scenes. Removing SCEM reduced the model's ability to handle cluttered backgrounds, as seen in the third row of Fig. 9, where the airplane's edge clarity diminished and one of the boxes was missed. This demonstrates SCEM's critical role in consolidating high-level feature representations, facilitating more accurate saliency predictions.

Overall, these ablation experiments demonstrate the critical roles of ASAM, DSFEM, and SCEM in LiteSalNet's architecture. Their individual contributions and synergistic interactions collectively enhance the model's ability to detect salient objects in complex and diverse remote sensing scenes.

2) Effectiveness of MSPDF: This section presents an ablation study of the MSPDF module, focusing on the roles of edge and skeleton streams in SOD. Several comparative experiments were conducted, as shown in Table III, to evaluate the contributions of these two streams.

The results in Table III demonstrate that incorporating additional streams improves performance across multiple evaluation metrics. Combining the salient object and edge streams improved F_β^{mean}, E_ξ^{mean}, and MAE, though S_m slightly decreased. This indicates that edge maps help refine object boundaries but may not fully capture internal structures.

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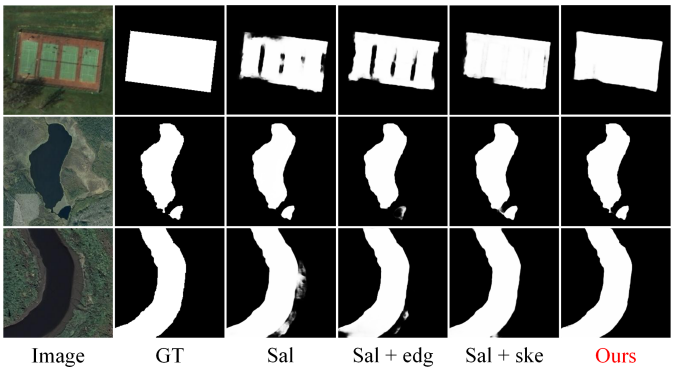

Fig. 10. Qualitative visualization of the effectiveness of salient edge and skeleton streams in the MSPDF module. Using only the salient object stream results in blurred boundaries and incomplete contours. Combining the salient object with the edge stream improves boundary sharpness but misses internal features while combining it with the skeleton stream preserves structure but weakens edge clarity. The full MSPDF configuration maintains both edge and structural details, demonstrating the complementary roles of edge and skeleton streams.

Integrating the salient object and skeleton streams resulted in consistent improvements across all metrics, highlighting the skeleton stream's role in preserving global object structures. Using all three streams (full MSPDF) yielded the best results, confirming the complementary roles of edge and skeleton streams. These findings demonstrate that combining structural guidance from both streams enhances detection accuracy and robustness.

Qualitative results in Fig. 10 further support these findings. Using only the salient object decoding stream led to blurred boundaries and indistinct contours, particularly in complex backgrounds and small-object scenarios. This highlights the need for structural guidance to preserve spatial integrity.

Combining the salient object and edge streams preserved some boundaries, though frequent structural errors occurred, especially for small or complex objects, emphasizing the skeleton stream's importance in maintaining structural consistency.

Similarly, integrating the salient object and skeleton streams maintained object structure but resulted in less distinct edges, particularly in cluttered backgrounds, reducing detection accuracy compared to the full configuration.

These experiments validate the essential roles of edge and skeleton streams within the MSPDF module. Edge streams ensure precise boundary localization, while skeleton streams maintain internal structure and spatial coherence. Their integration enhances model robustness and accuracy in challenging conditions, confirming the MSPDF module's design rationale and offering valuable insights for future model improvements.

3) Ablation Study and Component Analysis of the DSFEM Module: This section presents ablation experiments on the DSFEM module, focusing on the contributions of its key components: the Msaf, the Dmrf module, and the CSA mechanism. The results, presented in Table IV, highlight the impact of removing each module on four evaluation metrics: F_β^{mean}, E_ξ^{mean}, S_m, and MAE.

The results show that removing Msaf causes notable declines in F_β^{mean} and MAE, while slightly improving E_ξ^{mean} and S_m. This occurs because Msaf focuses on extracting multiscale features, especially for large objects and boundary details. Without Msaf, edge detection degrades, causing reduced precision and recall, thus lowering F_β^{mean}. However, since E-measure is less sensitive to local edge details, a slight improvement in global similarity is observed. Fig. 11 (first row) illustrates the model's failure to detect playground edges, confirming Msaf's importance in preserving boundary clarity.

TABLE IV
QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS EVALUATING THE CONTRIBUTIONS OF DIFFERENT COMPONENTS IN THE DSFEM MODULE

Components			EORSSD			
Msaf	**Dmrf**	**CSA**	$F_\beta^{\text{mean}} \uparrow$	$E_\xi^{\text{mean}} \uparrow$	$S_m \uparrow$	MAE $\downarrow$
✓	✓	✓	0.8809	0.9634	0.9273	0.0063
	✓	✓	0.8765	0.9637	0.9275	0.0066
✓		✓	0.8644	0.9586	0.9216	0.0067
✓	✓		0.8623	0.9684	0.9266	0.0073

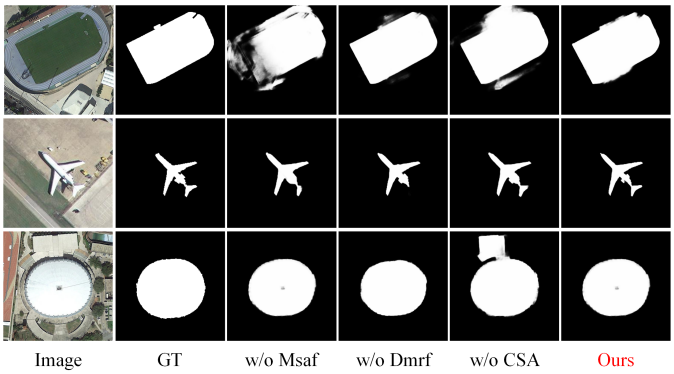

Fig. 11. Qualitative visualization of the impact of different components in the DSFEM module on SOD performance. Removing Msaf causes blurred boundaries and incomplete edge detection. Removing Dmrf results in missed small objects and loss of fine details, while removing CSA leads to fragmented object structures due to the lack of global contextual understanding. The full configuration effectively preserves edge details, structural integrity, and spatial consistency, demonstrating the complementary roles of these components.

Removing Dmrf results in a significant drop across all metrics, indicating its critical role in refining small-object features and enhancing low-contrast regions. Without Dmrf, the model struggles with fine-grained texture extraction, leading to missed objects and fragmented predictions. This negatively affects F_β^{mean} (due to lower recall), E_ξ^{mean} (due to reduced spatial coherence), and MAE (due to more pixel-level errors). As shown in Fig. 11 (second row), the model completely misses the airplane's tail, highlighting Dmrf's role in enhancing small-object detection and pixel-level accuracy.

Removing CSA significantly degrades F_β^{mean}, S_m, and MAE while slightly improving E_ξ^{mean}. This occurs because CSA establishes global contextual understanding by capturing long-range dependencies between features. Its removal causes incomplete object detection, reducing structural similarity (S_m) and recall, thereby lowering F_β^{mean}. However, since E-measure emphasizes overall spatial coherence rather than exact structure, E_ξ^{mean} temporarily improves. Fig. 11 (third row) shows salient object misidentification when CSA is

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absent, confirming its role in maintaining spatial consistency and structural integrity.

Combining Msaf, Dmrf, and CSA yields the best results across all metrics, confirming that these modules play complementary roles. Msaf focuses on large-scale object extraction and edge refinement, Dmrf targets small-object detection and low-contrast regions, while CSA ensures contextual coherence and global feature integration. Their integration enhances both object-level and pixel-level performance, validating the DSFEM module's design for robust saliency detection in complex scenarios.

V. Conclusion

This study introduces LiteSalNet, a lightweight network for SOD in optical remote sensing images. By integrating the ASAM, DSFEM, and SCEM, LiteSalNet efficiently refines multiscale features while leveraging the MSPDF for collaborative learning of saliency, edge, and skeleton features. This joint learning strategy enables LiteSalNet to achieve SOTA performance with only 3.90 M parameters and 7.35 G FLOPs, surpassing more complex models.

The balance between computational efficiency and performance is achieved through an optimized architectural design. ASAM, DSFEM, and SCEM enhance feature representations through multiscale feature fusion and attention mechanisms while maintaining a low parameter count. The MSPDF processes multitask features in parallel streams, reducing computational redundancy and boosting efficiency.

LiteSalNet's lightweight design and multitask processing capabilities suggest potential applications in areas, such as video analysis and multimodal learning. Future work could focus on extending LiteSalNet to dynamic environments by enhancing temporal feature extraction and integrating multimodal data, such as SAR, LiDAR, and multispectral imagery. These advancements would enable deeper semantic understanding and more precise saliency detection in complex remote sensing scenarios.

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